

Status — one line per numbered result

The short answer. PaperInventory.md is the long one and stays as the audit trail; this file is the thing to read first.

Measured on main after r0047: build **1364 jobs, 0 errors, 0 warnings**, sorry **0**, axiom **0**, sorryAx **0**. Package: 117 modules, 43,481 lines, 2,019 theorems.

Labels

Every paper property carries one of five labels. They answer two questions — **is it stated in Lean, and is it proved** — and the pair matters because **sorry can only see the second**.

#	Label	Stated?	Proved?	Meaning
1	S+P	yes, as the paper states it	yes	done
2	S+H	yes, as the paper states it	no — the proof is open	a real Lean statement with a hole. H is for <i>hypothesis</i> : the result is available only conditionally, as a theorem taking the open claim as an argument
3	S≠	yes, but not the paper's statement	yes	a weakening, an added hypothesis, or a deliberate repair of a printed defect
4	P	no — prose only	—	asserted in a docstring, never under the kernel
5	N	no statement at all	—	nothing in Lean mentions it

Two consequences worth stating once:

- **sorry 0 does not mean “everything is proved.”** A sorry is a hole in a proof someone *started*. Rows 4 and 5 were never started, and row 2's holes are carried as explicit hypotheses rather than as sorry, so none of the three appears in a sorry count. That is why this file reports rows 8–11 of the counts separately.
- **S≠ is not automatically a defect.** Five of the 18 are repairs of the paper's own printed errors, which is correct work; 13 are ours.

Counts

#		Count
1	Numbered results in Gunter & Scott	30
2	— fully proved	25
3	— partly proved, some part open	3
4	— refuted as stated (our transcription or the paper)	2
5	— status not yet established	1 (result 2; r0048 in flight)
6	Paper properties (numbered conjuncts + prose claims)	239
7	— S+P stated and proved	169
8	— S+H stated, proof open	12
9	— S≠ stated, but not as the paper states it	18 (13 ours, 5 repairs)
10	— P + N no Lean statement at all	36
11	Prop-valued claims nothing proves	7 (8 strict)

Rows 8 and 11 are the two kinds of unfinished work, and they are different: row 8 is a real Lean statement whose proof is open; row 11 is a def naming a claim **nobody attempted**. **sorry 0 sees neither** — which is why both are counted here and neither shows up in a build.

The 30 numbered results

P proved · p partly proved · R refuted as stated · — not in the tree

#	Result	Status
1	Theorem 1	P least fixed point, and below every fixed point
2	Theorem 2	— not quoted anywhere in the tree or the docs
3	Theorem 3	P theorem3, theorem3_existsUnique
4	Lemma 4	P NormalSubposet.lean
5	Lemma 5	P FinitaryProjection.lean
6	Theorem 6	P theorem6
7	Theorem 7	p sentence 1 proved; sentences 2–3 proved only at the degenerate strength — every domain has an EffectivePresentation because Classical.dec fills its fields. The recursive forms are open: Theorem7ArrowRecursive, Theorem7StrictRecursive
8	Lemma 8	P Product.lean
9	Lemma 9	P lem9_* — two printed misprints repaired (items 3 and 5)
10	Lemma 10	P 7 conjuncts, lem10_{prod, smash, sum, separated, lift, strict}
11	Theorem 11	P thm11, with thm11_converse
12	Theorem 12	P thm12 and the three powerdomains
13	Lemma 13	P lem13_smyth, lem13_hoare
14	Theorem 14	P thm14, both directions
15	Proposition 15	P “A bounded complete domain is bifinite” — printed folio 31. prop15 at Skeleton/Section6.lean:134, proved since that file was written. It is a Proposition, not a Theorem or Lemma , which is the whole reason it read as missing: both instruments that produced the gap searched only for thm theorem lem lemma
16	Theorem 16	P thm16, thm16_positive
17	Lemma 17	P 10 conjuncts. r0047 removed [BoundedComplete β] from lem17_fun and lem17_strictFun
18	Theorem 18	P thm18 — closed r0042, [propext, Classical.choice, Quot.sound]
19	Lemma 19	P lem19
20	Lemma 20	P lem20
21	Theorem 21	P thm21
22	Theorem 22	P thm22
23	Lemma 23	P lem23
24	Lemma 24	P lem24 (Gunter & Scott’s; not Gunter 1987’s Lemma 24, below)
25	Theorem 25	P thm25, thm25_isUniversal
26	Theorem 26	p thm26 carries an added $hs : \forall i, 0 < s_i$ — no nullary operations — which the paper does not assume. The recorded justification, that Theorem 26 is false for a signature admitting arity 0, is not established : the contradiction is derived from $fst(\psi(x)) = x$, a property of the paper’s <i>construction</i> and of our own conclusion, whereas the printed statement asks only that A “be made isomorphic to a subalgebra” — and two one-point algebras are isomorphic to the same one-point subalgebra $\{F_1\}$. The argument refutes the paper’s proof at arity 0, not its theorem. So hs is a defect of ours, not a repair. To settle it: exhibit a genuine arity-0 counterexample to the <i>printed</i> conclusion, or drop hs and prove thm26 without it
27	Theorem 27	P thm27
28	Lemma 28	R refuted at generic U (not_forall_lemma28, witness Flat Empty), and stays false after adding [Domain U] and [BoundedComplete U]. Proved at U : lemma28AtU, all nine conjuncts. The blocker is UniversalForBCD U
29	Theorem 29	p sentence 1 proved (thm29). Sentence 2: Thm29Second refuted — our transcription dropped the paper’s word “domain”. Thm29SecondAtDomains, the true reading, is open
30	Lemma 30	R universal closure refuted (not_forall_lemma30). Lemma30AtV is open , now at arity 3

Result 15 was never missing — the instruments were. Both the declaration scan and the docstring grep searched only for thm|theorem|lem|lemma, and Gunter & Scott number **Propositions in the same sequence**. scripts/numbered-status.sh now matches prop|proposition|cor|corollary as well; the widening was measured before it was made, and over results 1–30 it adds exactly one hit and no false positive.

Result 2’s status is not yet established and is being checked against the printed text. Results 4, 5 and 8 carry no numbered declaration either, but are each quoted in a module docstring — NormalSubposet.lean, FinitaryProjection.lean, Product.lean — so **a result stated under a descriptive name is the normal case here**, and a missing number is not evidence of a missing proof.

The figure “30 numbered results” is still **unverified** against the printed text. It comes from `PropertiesVsTheorems.md` and no round has checked it.

What is open, in one table

#	Item	What it needs
1	StepFunctionsDecidable	restate over consistentEnum — the guard <code>IsCompactElement (ofPairs Q)</code> is <i>not</i> the boundedness test, kernel-checked. Machinery built. Closes 4 claims
2	Theorem7ArrowRecursive	item 1, then <code>Primrec</code> facts for the <code>Finset (ℕ × ℕ)</code> coding
3	Theorem7StrictRecursive	item 1, likewise
4	Thm29Normal	the <code>M(f)</code> tower refutes Theorem 25’s hypothesis; the η tower is not a fixed point. Three routes named, one is refuting it outright
5	Thm29SecondAtDomains	item 4
6	Lemma30AtV	item 4, plus <code>FpImagesBifinite V</code>
7	Lem30Arrow	item 6
8	PreservesRecursivePresentation	proved at <code>fst0p/snd0p</code> ; open at the arrow, where it is equivalent to item 2

Items 2–3 and 5–7 are downstream of 1 and 4. Only items 1 and 4 are real work, and item 1 is mechanical.

Additional theories

Results from other authors, formalized here because the paper’s proofs need them. None is part of Gunter & Scott’s 30.

Jung, *Cartesian Closed Categories of Domains*

#	Result	Status
1	Corollary 1.36	proved — <code>JungCor136</code> , not by Jung’s route (which goes through <code>Prop. 1.22</code> and a retraction pair) but by indexing below <code>cap e</code>
2	Theorem 1.37	proved at [Domain D] — <code>R45.Agent5.thm137</code> . Full generality open
3	Theorem 1.37 for chains	proved at [Domain D] — <code>thm137Chains</code>
4	Proposition 1.22	not needed — the route around it is item 1
5	Theorem 2.1, 2.3	quoted; five printed defects in Jung’s write-up recorded

Iwamura and Markowsky

#	Result	Status
1	Iwamura’s lemma	proved — <code>exists_chain_directed_cover</code> . Never in Mathlib
2	Markowsky’s theorem	proved — <code>hasChainSuprema_iff_hasDirectedSuprema</code> . Never in Mathlib
3	<code>HasWellOrderedInfima</code>	no producer exists — five occurrences, all consumers. A complete reduction chain with an empty left end

Spreen 2005

#	Result	Status
1	Lemma 5.8	proved — <code>PropertyM.hasOmega0pBoundsAbove_pair</code> . This, not Iwamura, is what closed Jung’s 1.37 here

Gunter 1987, *Universal Profinite Domains*

#	Result	Status
1	Lemma 24 at $M(A)$	proved — <code>R47.Agent1.lemma24_MPair</code> . The first proof anywhere: Gunter’s printed Lemma 24 produces <i>some</i> A^+ and is not about $M(A)$; the $M(A)$ form is a remark with no theorem and no proof
2	<code>HasNormalRealizations</code> at <code>Ainf</code>	refuted — <code>not_hasNormalRealizations_Ainf</code> . The route to <code>Thm29Normal</code> is sound but its hypothesis is unsatisfiable at this tower
3	Proposition 21, Theorems 22 and 25	used, via the implication <code>thm29Normal_of_hasNormalRealizations</code>

Defects in the printed paper

Nine, recorded in `StatementRecovery.md`. Three suspected tenths were checked and turned out to be **our** transcription errors, not the paper’s:

#	Suspected	Actually
1	Theorem 26 false at arity 0	refutes the paper’s <i>proof</i> , not its statement
2	Theorem 29’s second sentence false	our transcription dropped “domain” from “any bifinite domain”
3	the step-function enumeration	our guard tests compactness where the paper tests boundedness

Reproducing

```

scripts/counts.sh           # modules, lines, theorems, sorry
scripts/numbered-status.sh  # declarations per numbered result
scripts/a6-env-scan.sh <out> # then a6-summarize.py for row 11
scripts/a5-r47-conditional.sh # row 8, the S+H conditional surface
scripts/compile.sh -r <round> # build

```